Section 3.3

29. a. $E(X)=\sum_{x\in D} x\,p(x)$

$=(1)(.05)+(2)(.10)+(4)(.35)+(8)(.40)+(16)(.10)$

$=6.45$ GB

b. $V(X)=\sum_{x\in D}(x-\mu)^2\cdot p(x)$

$=(1-6.45)^2\cdot 0.05+(2-6.45)^2\cdot 0.10+\cdots+(16-6.45)^2\cdot 0.10$

$=15.6475\ (\text{GB})^2$

c. $\sigma=\sqrt{V(X)}=\sqrt{15.6475}=3.957$ GB

d. $V(X)=E(X^2)-[E(X)]^2$

$E(X^2)=1^2\times 0.05+2^2\times 0.10+\cdots+16^2\times 0.10$

$=57.25$

$V(X)=57.25-6.45^2=15.6475\ (\text{GB})^2$

33. a. $E(X^2)=1^2\cdot p+0^2\cdot(1-p)=p$

b. $E(X)=p$. Thus $V(X)=E(X^2)-[E(X)]^2=p-p^2=p(1-p)$

c. $E(X^{79})=1^{79}\cdot p+0^{79}\cdot(1-p)=p$

38. $E(h(X)) = E(1/X) = \sum_{i=1}^{6} \frac{1}{i} \cdot \frac{1}{6} = 0.408$

Since $\frac{1}{3.5} = 0.286 < 0.408$

I will gamble.

41. $V(aX+b) = \sum (ax+b - E(aX+b))^2 \cdot p(x)$

Since $E(aX+b) = a\mu + b$.

$V(aX+b) = \sum (ax - a\mu)^2 \cdot p(x)$

$= a^2 \sum (x-\mu)^2 \cdot p(x)$

$= a^2 \cdot \sigma_X^2$

Section 3.4

46. a. $b(3;8,.35) = \binom{8}{3}\cdot(.35)^3(1-.35)^{8-3} = 0.279$

b. $b(5;8,.6) = \binom{8}{5}\cdot(.6)^5(1-.6)^{8-5} = 0.279$

c. $P(3 \le X \le 5) = b(3;7,.6) + b(4;7,.6) + b(5;7,.6)$

$= \binom{7}{3}\cdot(.6)^3(.4)^4 + \binom{7}{4}\cdot(.6)^4(.4)^3 + \binom{7}{5}\cdot(.6)^5(.4)^2$

$= 0.745$

d. $P(1 \le X) = 1 - P(X=0)$

$= 1 - b(0;9,.1)$

$= 1 - \binom{9}{0}\cdot(.1)^0(.9)^9$

$= 0.613$

47. a. $B(4;15,.3) = 0.515$

b. $b(4;15,.3) = B(4;15,.3) - B(3;15,.3) = 0.219$

c. $b(6;15,.7) = B(6;15,.7) - B(5;15,.7) = 0.012$

d. $P(2 \le X \le 4) = B(4;15,.3) - B(1;15,.3) = 0.480$

e. $P(2 \le X) = 1 - B(1;15,.3) = 0.965$

f. $P(X \le 1) = B(1;15,.7) = 0.000$

g. $P(2 < X < 6) = B(5;15,.3) - B(2;15,.3) = 0.595$

48. a. $P(X\le 2)=B(2;25,0.05)=0.873$

b. $P(X\ge 5)=1-P(X\le 4)=1-B(4;25,0.05)=1-0.993=0.007$

c. $P(1\le X\le 4)=B(4;25,0.05)-B(1;25,0.05)=0.480$

d. That is, $P(X=0)=b(0;25,0.05)=B(0;25,0.05)=0.277$

e. $E(X)=np=25\times 0.05=1.25$

$V(X)=\sigma^2=np(1-p)=1.25\times 0.95=1.1875$

Thus standard derivation $\sigma=\sqrt{V(X)}=1.09$

54. a. The probability satisfies the binomial distribution. $n=10$. $p=0.6$.

Let X = the number of customers that want the oversize version. We want to find.

$P(X\ge 6)=1-P(X\le 5)=1-B(5;10,0.6)=1-0.367=0.633$

b. $E(X)=np=6$ $V(X)=np(1-p)=2.4$. $\sigma=\sqrt{V(X)}=\sqrt{2.4}=1.55$

Thus $E(X)-\sigma=4.45$, $E(X)+\sigma=7.55$.

Thus we want to find

$P(5\le X\le 7)=B(7;10,0.6)-B(4;10,0.6)=0.667$

c. We want to find $P(3\le X\le 7)$ since each version has at most 7 rackets.

$P(3\le X\le 7)=B(7;10,0.6)-B(2;10,0.6)=0.821$